

Assignment 4

Neural Style Transfer

Introduction

In this assignment, I implemented Neural Style Transfer using a pretrained deep learning model called VGG19. The goal of this task is to combine two images - a content image and a style image. The final output image keeps the structure of the content image but looks like the style image

How the Method Works

The model used in this assignment is VGG19, which is a pretrained convolutional neural network. We only used the feature extraction part of the model, not the classification part.

The idea is:

- Pass images through the network
- Extract features from certain layers
- Compare those features to calculate loss

Content and Style Representation

Content

The content of the image is taken from a deeper layer of the network (conv4_2).

This layer captures the structure of the image like shapes and objects.

The content loss is calculated by comparing:

- target image features
- content image features

Style

The style is taken from multiple layers:

- conv1_1
- conv2_1
- conv3_1
- conv4_1
- conv5_1

To represent style, we use something called a Gram Matrix, which captures textures and patterns.

The style loss measures how different the style of the target image is from the style image.

Total Loss

We combine both losses:

- Content Loss - keeps structure
- Style Loss - adds artistic style

The total loss is:

Total Loss = Content Loss + Style Loss

Training Process

We start with a copy of the content image as the target image.

Then:

- Pass the target image through VGG19
- Compute content and style loss
- Update the target image using an optimizer (Adam)
- Repeat for many steps

Over time the image slowly changes and becomes stylized.

Experiments

I tried different hyperparameters to see how results change.

Experiment 1:

Learning rate: 0.003

Steps: 2000

Result:

Good quality output but very slow, especially on CPU.

Experiment 2:

Learning rate: 0.01

Steps: 800

Result:

Faster and still good quality. This was the best setting.

Experiment 3:

Style weight reduced

Result:

Less artistic effect. The image looked more like the original content.

Observations

- Higher style weight - more artistic effect
- Lower style weight - more original content
- More steps - better quality but slower
- GPU (CUDA) makes training much faster

Challenges Faced

- I got an error with conv4_2 because of incorrect layer mapping
- Running on CPU was very slow
- Needed to make sure all cells run in correct order

Conclusion

In this assignment, neural style transfer using VGG19. The model was able to combine content and style from two images and generate a new artistic image.

Screenshot:



Steps: 800
Learning rate: 0.01
Step 200/800, Total loss: 14586266.0000



Step 400/800, Total loss: 14477612.0000



Step 800/800, Total loss: 14394962.0000



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